

# Eye Wonder

## Team 4 (Nicognicogni) Occurrence 2

|                                                                              |                                                                                |                                                                                 |
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### ABSTRACT

Cluttered, everyday environments fundamentally change how we locate objects, yet why searching efficiency varies so heavily between different individuals and scene layouts remains poorly understood. To explore this dynamic, we deployed a screen-based Tobii eye tracker to record the gaze behaviours of 130 Universiti Malaya's undergraduates across 13 naturalistic stimulus images. Each scene contained manually delineated Target and Distractor Areas of Interest (AOIs). After clearing out testing logs and duplicated records, our final analytics pipeline processed 7150 distinct observation rows. The statistical results highlighted a massive performance gap between target and distractor processing. Distractor AOIs consistently grabbed early attention much faster than the actual search objectives (Distractor  $M = 3434.87$  ms, Target  $M = 5611.36$  ms;  $t(129) = 16.761$ ,  $p < 0.001$ , Cohen's  $d_z = 1.470$ ). However, once participants successfully locked onto a target, they fixated on it significantly longer to cognitively verify the item (Target  $M = 871.74$  ms, Distractor  $M = 660.64$  ms;  $t(129) = 7.739$ ,  $p < 0.001$ , Cohen's  $d_z = 0.679$ ). A Friedman test confirmed that the background visual chaos heavily dictate overall search difficulty ( $\chi^2(12) = 625.232$ ,  $p < 0.001$ ). Looking closer at individual

variations, K-means clustering successfully mapped out two distinct behavioral profiles: 76 Fast Searchers and 54 Slow Searchers, which a Random Forest model later confirmed with a 87% predictive accuracy. Ultimately, these results provide solid, real-world backing for Feature Integration Theory, proving that while bottom-up cues drive a fast distractor capture, top-down validation rules target verification.

Keywords : Eye-Tracking, Visual Search, Feature Integration Theory, K-means Clustering, Random Forest

## 1. INTRODUCTION

### 1.1 Study Background

Feature Integration Theory (FIT) explains that visual search in complex environments is guided by environmental cues, distinguishing between rapid parallel searches and slower serial searches. Eye-tracking technology quantifies this attention using metrics such as total fixation duration and Time to First Fixation (TTFF). In a study of 130 participants viewing 13 images with Target and Distractor Areas of Interest (AOIs), data revealed that while distractors captured early attention faster, participants fixated significantly longer on targets to cognitively verify them.

### 1.2 Research Gap

The research identifies a significant limitation in existing eye-tracking studies, which frequently utilize artificial stimuli that do not accurately represent complex and real-world environments. Furthermore, traditional group-level analysis often overlooks major individual variations in search efficiency and scanning behavior. To address these gaps, the Eye Wonder study employs ecologically valid scenes and unsupervised machine learning to analyze the behavioral variations of 130 participants, comparing fixation metrics across different stimulus images and Areas of Interest (AOI).

### 1.3 Research Questions

**RQ1:** How does visual attention differ between Target AOIs and Distractor AOIs during visual search tasks?

**RQ2:** Do different stimulus images produce different levels of visual search difficulty as reflected by target total fixation duration?

**RQ3:** Can participants be grouped into distinct visual search behaviour profiles based on eye-tracking metrics?

## 1.4 Research Objectives

| Obj        | Specific                                                                                                                 | Measurable                                                                             | Achievable                                                                           | Relevant                                                                               | Time-bound                 |
|------------|--------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|----------------------------|
| <b>RO1</b> | To compare visual attention between Target AOs and Distractor AOs using total fixation duration and TTFF.                | Paired samples t-test on total fixation duration and TTFF; Cohen's dz for effect size. | Cleaned dataset of 130 participants ready; Python libraries available.               | Directly tests FIT prediction that fixation concentrates on the search target.         | Weeks 9–12 analysis phase. |
| <b>RO2</b> | To determine whether visual search difficulty differs across 13 stimuli using target total fixation duration.            | Friedman test on target total fixation duration across the 13 stimulus images.         | Multi-stimulus dataset available; standard statistical tests applicable.             | Quantifies how stimulus characteristics influence search efficiency.                   | Weeks 9–12 analysis phase. |
| <b>RO3</b> | To identify distinct visual search behaviour profiles among participants using clustering based on eye-tracking metrics. | K-Means clustering ( $k = 2$ ) on fixation duration, TTFF, and fixation count.         | K-Means feasible for $N = 130$ ; scikit-learn available in the analysis environment. | Reveals individual-level behavioural variation not captured by group-level statistics. | Weeks 9–12 analysis phase. |

Table 1: Research Objectives

## 1.5 Research Hypotheses

|                  | Null Hypothesis ( $H_0$ )                                                                                     | Alternative Hypothesis ( $H_1$ )                                                                             | Statistical Test                                                         |
|------------------|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| <b>H01 / H11</b> | There is no significant difference in total fixation duration and TTFF between Target AOs and Distractor AOs. | There is a significant difference in total fixation duration and TTFF between Target AOs and Distractor AOs. | Paired samples t-test; reject $H_0$ if $p < 0.05$ ; Cohen's dz reported. |
| <b>H02 / H12</b> | There is no significant difference in target total fixation duration across the 13 stimulus images.           | There is a significant difference in target total fixation duration across the 13 stimulus images.           | Friedman test; reject $H_0$ if $p < 0.05$ .                              |

|                          |                                                |                                                                                              |                                                                                  |
|--------------------------|------------------------------------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <b>RO3 (Exploratory)</b> | No formal null hypothesis, RO3 is exploratory. | K-Means clustering will reveal distinct visual search behaviour profiles among participants. | K-Means clustering (k = 2) based on fixation duration, TTFF, and fixation count. |
|--------------------------|------------------------------------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|

Table 2: Research Hypotheses

## 1.6 Contributions

| Type of Contribution  | Description                                                                                                                                                                                                                  |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Theoretical</b>    | Offers empirical evidence on visual attention, enhancing the understanding of target/distractor differentiation and how stimulus features influence search behavior.                                                         |
| <b>Methodological</b> | Validates total fixation duration and TTFF as visual attention indicators. Combining descriptive statistics, inferential tests, and clustering provides a replicable eye-tracking research framework.                        |
| <b>Practical</b>      | Results can guide the development of interfaces and displays by detailing gaze allocation and factors impacting target fixation and search efficiency.                                                                       |
| <b>Data Analytics</b> | Illustrates the application of K-Means clustering to eye-tracking data, showing how machine learning can uncover behavioural patterns, such as fast vs. slow searcher subgroups, that are invisible in aggregate statistics. |

Table 3: Contributions

## 2. LITERATURE REVIEW

### 2.1 Feature Integration Theory

Visual search is guided by the distinction between parallel and serial processing, as defined in Feature Integration Theory (FIT). Treisman and Gelade (1980) proposed that basic features like colour and motion are processed automatically in a pre-attentive stage. When targets resemble distractors, a slower focused attention stage is necessary for feature binding. FIT predicts that salient distractors capture early gaze, while targets require longer verification fixations. This study evaluates these predictions by comparing Time to First Fixation (TTFF) and Total Fixation Duration (TFD) between target and distractor Areas of Interest (AOIs).

### 2.2 Eye-Tracking Metrics and Visual Search

Eye-tracking provides objective quantification of attention through metrics such as Total Fixation Duration (TFD), Time to First Fixation (TTFF), and fixation count. TFD reflects cumulative cognitive processing on a region, while TTFF indicates how rapidly a region captures attention. A meta-analysis by Mann et al. (2007) found that higher expertise in visual tasks is consistently associated with decreased TTFF, though fixation duration

findings vary by task complexity. Critically, Wolfe et al. (2011) demonstrated that most eye-tracking studies rely on artificial stimuli such as geometric shapes on blank backgrounds, limiting generalisability to real-world scenes. The present study addresses this gap by using 13 naturalistic stimulus images with embedded targets.

### **2.3 Stimulus Characteristics and Search Difficulty**

Search difficulty varies systematically with stimulus properties. Ernst and Wolfe (2022) manipulated distractor heterogeneity, target-distractor similarity, and stimulus density, finding that fixation duration becomes a more sensitive index of search difficulty in complex scenes compared to simple displays. This supports the use of naturalistic stimuli in the present study to examine how different images (RQ2) produce variation in target fixation duration.

### **2.4 Individual Differences in Visual Search**

Hayes and Henderson (2017) showed that scan patterns during scene viewing predict individual differences in cognitive capacity. More directly, Hsiao et al. (2021) applied complex co-clustering frameworks to eye-movement data and discovered two distinct patterns among participants: explorative (switching between regions) and focused (concentrated on the foreground). While complex temporal patterns have historically required hidden Markov modelling, coarser global variations in speed and accuracy can be efficiently mapped using unsupervised partitioning. Furthermore, Peltier and Becker (2020) demonstrated that fluid intelligence and working memory capacity account for significant variance in search accuracy. These findings motivate the exploratory K-means clustering approach in the present study (RQ3) to identify distinct searcher profiles.

### **2.5 Summary of Research Gaps**

Three gaps guide the present study. First, limited ecological validity exists in most eye-tracking research due to artificial stimuli. Second, individual-level behavioural variation remains underexplored. Third, few studies systematically compare fixation metrics across multiple distinct stimulus images. The present study addresses each gap sequentially through its three research questions.

## **3. METHODOLOGY**

### **3.1 Participants**

A total of 130 students from the Faculty of Computer Science and Information Technology of the University of Malaya were recruited for this study. The participants

were students enrolled in WID2003 Cognitive Science. There was no financial reward offered and participants were instructed about the nature of the visual search task.

### 3.2 Materials

#### Hardware

The Tobii Eye Tracker was built into a desktop workstation in the BT2 tutorial room to record participants eye movements. The apparatus used a corneal reflection tracking technology using near infrared illumination of the eye and calculation of the gaze coordinates using the relative displacement of the pupil centre and the corneal reflection. The advantage of this technique is that it allows for accurate non-invasive spatial localization of fixations without the need of any head mount device.

#### Software

Tobii Pro Lab was used for the stimulus presentation and real-time gaze recording and generation of post hoc metrics. The software enabled the automated detection of events, real-time monitoring of the tracking quality during the recording session, and the calculation of eye-tracking metrics within researcher-defined Areas of Interest (AOIs).

#### Stimuli

The stimulus set consisted of 13 cartoony pictures ranging from pictures of families eating dinner to the scenery of forest. The target object was embedded in a series of pictures with a number of visually competing objects.

Two categories of spatial AOI envelopes were manually created for each stimulus image using the polygon drawing tool in Tobii Pro Lab:

- **Target AOI** : Spatial areas that surround the Target of Interest of each scene.
- **Distractor AOIs**: Spatial areas that do not surround the Target of Interest but meaningful enough that there is a lot of fixation in that particular area.

The research team drew AOI boundaries to achieve exact spatial alignment of the contours of objects with the AOI to allow gaze events to be attributed to the Target AOI and Distractor AOI regions during analysis.

### 3.3 Procedure

This task was performed in a controlled environment in a room where participants sat around 60–65cm from an eye-tracking monitor. To allow for the accuracy to be established, calibration of the eye tracker was done before the task by having the participants track a moving dot on the screen. After calibration, participants read the instructions displayed on screen before going on to the 13 formal visual search tasks.

The image was presented for 15 seconds, during which the participants searched for the designated target object and gaze data was collected by the device.

## 4. ANALYSIS

### 4.1 Data Preprocessing

The raw eye-tracking data were exported from Tobii Pro Lab as a tab-separated file, containing 14,740 rows and 54 columns from 131 recorded participants. Data preprocessing was carried out in Python using pandas. Rows that were not part of the main task, such as instruction screens and practice trials, were removed by filtering the dataset to the 13 task stimuli only. The test participant ('testing1') and an earlier duplicate session for participant VT38 were also excluded. After cleaning, the final dataset consisted of 7,150 rows from 130 participants, covering 13 stimuli and 55 AOI rows per participant. Missing TTFF values were kept as NaN because they indicated AOIs that participants never fixated on, which represents a meaningful behavioural outcome rather than a data error. The cleaned dataset was saved as Visual\_Search\_Cleaned.csv, and the full preprocessing pipeline is available in here: [eye\\_wonder\\_analysis.ipynb](#).

### 4.2 Eye-Tracking Features and Descriptive Analysis

For each participant–AOI combination, **three eye-tracking features** were extracted: **Total Fixation Duration (TFD)**, **Time to First Fixation (TTFF)**, and **Number of Fixations**. TFD refers to the total amount of time a participant spent looking at an AOI, while TTFF indicates how quickly that AOI was first noticed. Number of Fixations shows how often participants returned to the same AOI. Together, these measures capture different aspects of visual attention: TFD reflects processing depth, TTFF reflects attentional capture, and fixation count reflects re-examination behaviour.

Before conducting the inferential tests, the data were first visualised to identify the main patterns. Figure 1 presents the mean values of the three metrics by AOI type, comparing Target and Distractor AOIs. Figure 2 shows the mean target fixation duration across the 13 stimuli, with SD error bars, while Figure 3 presents the mean TTFF for each stimulus. Figure 4 provides a heatmap of mean fixation counts across stimuli and AOI types. Overall, these visualisations show that Target AOIs generally received longer fixation durations, whereas Distractor AOIs tended to attract earlier and more frequent gaze. They also show clear differences in search difficulty across the stimuli. These patterns provided the basis for the inferential analyses reported in the next section.

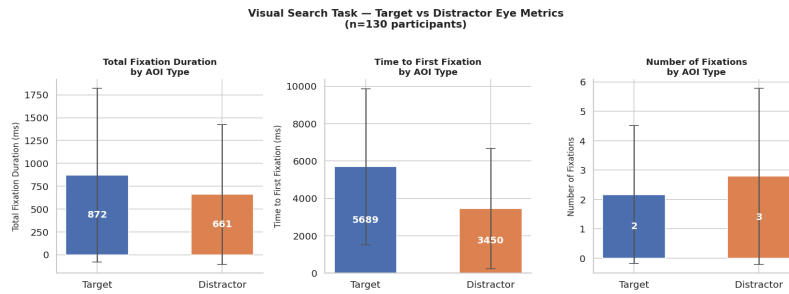


Figure 1: Target vs Distractor eye metrics

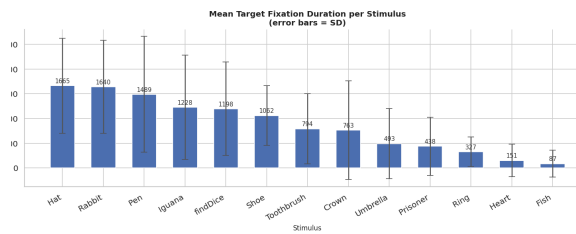


Figure 2: Mean target fixation duration per stimulus

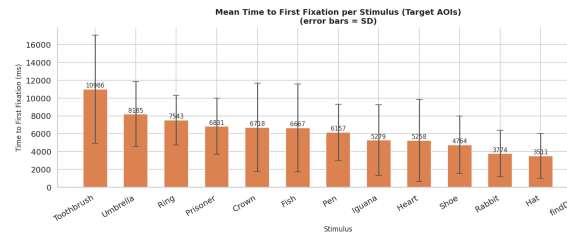


Figure 3: Mean TTFF per stimulus

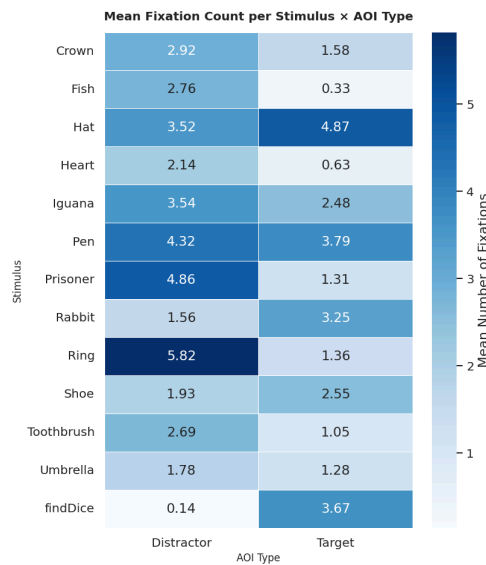


Figure 4: Fixation Count Heatmap (Stimulus x AOI Type)

### 4.3 Inferential Statistical Analysis

Two inferential tests were used to address RQ1 and RQ2.



For RQ1, **paired samples t-tests** were chosen because the same 130 participants contributed measurements for both Target and Distractor AOIs. For each participant, the mean Target and Distractor values were calculated and compared directly. Separate t-tests were conducted for TFD and TTFF, with Cohen's  $d_z$  reported as the effect size for the within-subject comparison.

For RQ2, a **Friedman test** was used because all 130 participants viewed the same 13 stimuli, creating a repeated-measures design. Since eye-tracking measures such as TFD are often not normally distributed, the non-parametric Friedman test was considered more appropriate than a repeated-measures ANOVA. This test compared participant-level mean target TFD across the 13 stimuli.

#### 4.4 Advanced Analysis Method

To address RQ3, two machine learning approaches were used.

For the **K-Means clustering analysis** ( $k = 2$ ), participant-level features were first prepared using mean TFD, mean TTFF, and mean number of fixations from Target AOIs only. These features were standardized using z-scores so that no single metric would dominate the clustering because of scale differences. A two-cluster solution was chosen to reflect the FIT-based distinction between more efficient and less efficient searchers. After examining the cluster means, the groups were labelled as Fast Searchers, who showed lower TTFF values, and Slow Searchers, who showed higher TTFF values.

To add another view of participants' scanning behaviour, an **AOI transition matrix** was also created. The sequence of AOI visits was reconstructed from the raw data, and the matrix showed how often gaze shifted between different AOI pairs across participants and stimuli. This helped capture search strategy in a way that went beyond the average fixation measures used in the clustering analysis.

Finally, a **Random Forest classifier** was used to test whether the Fast/Slow Searcher profiles could be recovered from the same three eye-tracking features. The cluster labels were treated as the target variable, and the dataset was split into 70% training data and 30% testing data. Model performance was evaluated using accuracy and F1-score, while feature importance was used to identify which eye-tracking metric contributed most to the classification.

## 5. RESULTS

### 5.1 Descriptive Overview

The descriptive charts in Section 4.2 (Figures 1–4) reveal a consistent pattern across all 130 participants. Target AOIs received longer total fixation durations on average, whereas Distractor AOIs were reached earlier (lower TTFF) and revisited more frequently (higher fixation count). This suggests that once participants identified the target, they spent more time verifying it, while distractors captured initial gaze more readily. Variation across the 13 stimuli was also apparent in Figures 2 and 3, with some stimuli producing notably longer fixation durations and TTFF than others, indicating differences in search difficulty.

### 5.2 Inferential Statistics (RQ1 and RQ2)

**RQ1 — Total Fixation Duration:** Participants spent significantly more time fixating on Target AOIs ( $M = 871.74$  ms,  $SD = 338.04$ ) than on Distractor AOIs ( $M = 660.64$  ms,  $SD = 155.36$ ),  $t(129) = 7.739$ ,  $p < 0.001$ , Cohen's  $d_z = 0.679$ . This moderate-to-large effect suggests that once the target was located, participants engaged in deeper visual and cognitive processing of that region.

**RQ1 — Time to First Fixation:** TTFF was also significantly longer for Target AOIs ( $M = 5611.36$  ms,  $SD = 1605.18$ ) than for Distractor AOIs ( $M = 3434.87$  ms,  $SD = 747.72$ ),  $t(129) = 16.761$ ,  $p < 0.001$ , Cohen's  $d_z = 1.470$ . This very large effect indicates that distractors captured participants' gaze much earlier, with targets being fixated approximately 2,176 ms later on average.

**RQ2 — Stimulus Comparison:** The Friedman test showed a significant difference in participant-level mean target fixation duration across the 13 stimulus images,  $\chi^2(12) = 625.232$ ,  $p < 0.001$ . This finding indicates that visual search difficulty varied substantially between stimuli, likely due to differences in layout complexity and target–distractor similarity.

Table 4 summarises the inferential results. Figure 5 further supports these findings through boxplots of participant-level Target versus Distractor TFD in the left panel and per-stimulus target TFD in the right panel.

| Metric / Test                                           | Target M<br>(SD)     | Distractor M<br>(SD) | Test Statistic   | p-value   | Effect Size<br>(Cohen's $d_z$ ) |
|---------------------------------------------------------|----------------------|----------------------|------------------|-----------|---------------------------------|
| <b>Total Fixation Duration</b> ( <i>Paired t-test</i> ) | 871.74ms<br>(338.04) | 660.64ms<br>(155.36) | $t(129) = 7.739$ | $< 0.001$ | 0.679                           |

|                                                      |                        |                       |                        |           |       |
|------------------------------------------------------|------------------------|-----------------------|------------------------|-----------|-------|
| <b>Time to First Fixation (TTFF) (Paired t-test)</b> | 5611.36ms<br>(1605.18) | 3434.87ms<br>(747.72) | $t(129) = 16.761$      | $< 0.001$ | 1.470 |
| <b>Across 13 Stimuli (Friedman test)</b>             | Varies                 | Varies                | $\chi^2(12) = 625.232$ | $< 0.001$ | N/A   |

Table 4: Statistical Results

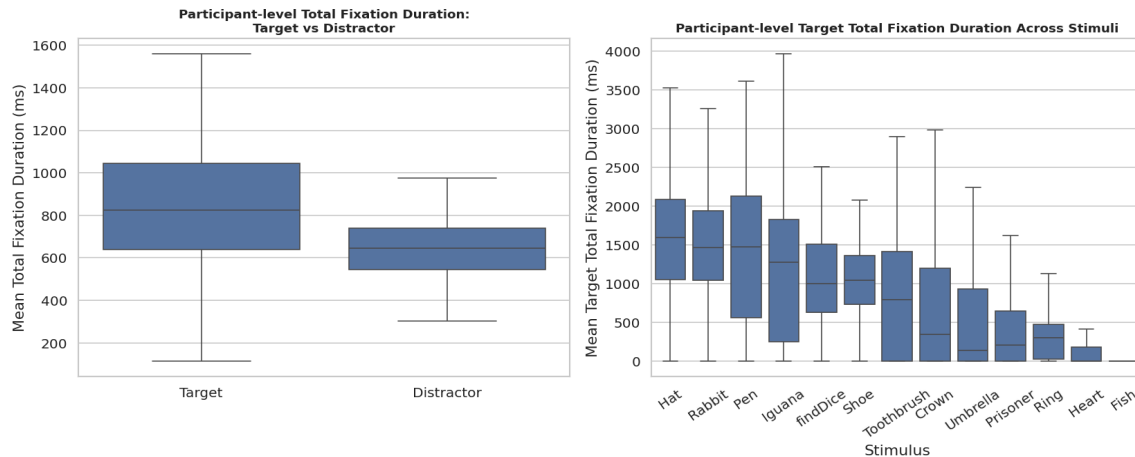


Figure 5: Boxplot of Target vs Distractor TFD and per-stimulus TFD

### 5.3 Advanced Analysis Results (RQ3)

**K-Means Clustering:** The K-Means analysis ( $k = 2$ ) identified two clear visual search profiles among the 130 participants: 76 Fast Searchers and 54 Slow Searchers. Fast Searchers had lower mean TTFF values, suggesting that they located the target more quickly. In contrast, Slow Searchers showed higher TTFF values and longer fixation durations, indicating a more effortful search process. As shown in Figure 6, the clustering scatter plot separates the two groups clearly, mainly along the TTFF axis.

**Transition Matrix:** The AOI transition matrix in Figure 7 shows how often participants shifted their gaze between target and distractor regions. Brighter cells represent more frequent transitions between specific AOI pairs. Overall, the pattern suggests that participants often moved their gaze between distractor AOIs before eventually settling on the target. This is consistent with a serial search strategy, especially for the more difficult stimuli.

**Random Forest Classification:** A Random Forest model was trained using three participant-level features: mean TFD, mean TTFF, and mean fixation count. The model predicted Fast/Slow Searcher labels with 87% accuracy on the held-out test set ( $n = 39$ ). The F1-scores were 0.88 for Fast Searchers and 0.86 for Slow Searchers. As shown in Figure 8, mean fixation duration contributed the most to prediction (importance = 0.39), followed by mean TTFF (0.34) and mean number of fixations (0.27). This strong

performance suggests that the cluster labels reflected meaningful behavioural differences rather than arbitrary groupings.

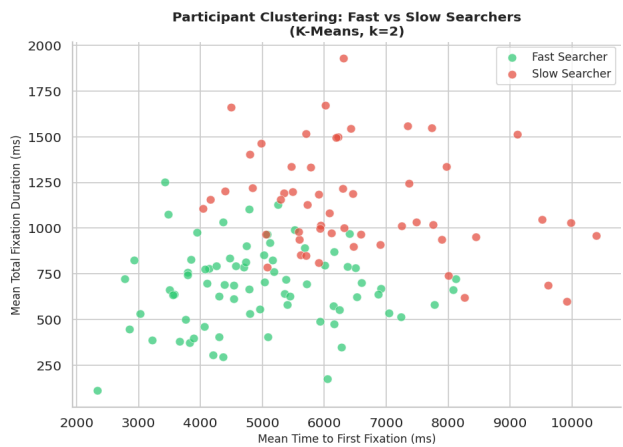


Figure 6: K-Means clustering scatter plot

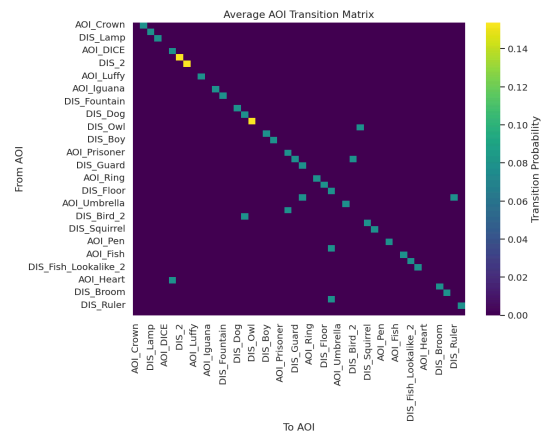


Figure 7: Transition matrix

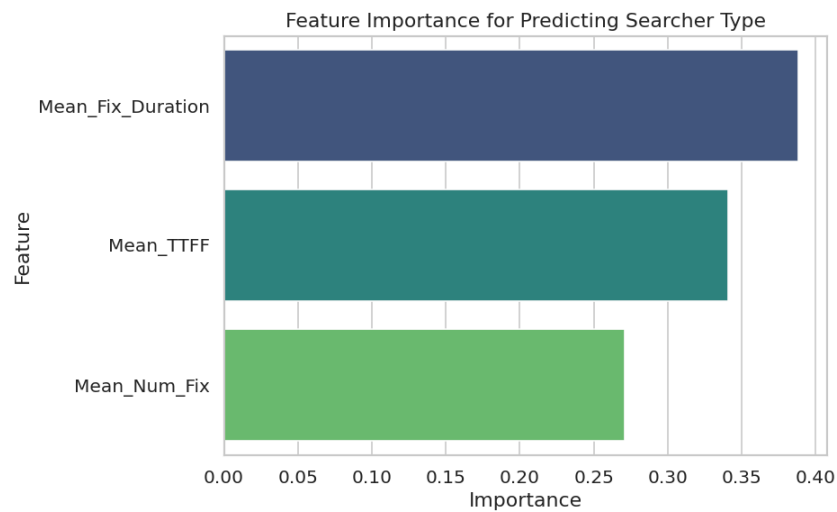


Figure 8: Feature importance for Predicting Searcher Type

6. DISCUSSION

Participants distributed visual attention unequally between target and distractor regions. Distractor AOIs had lower TTFF values, indicating they were often fixated on before the target. This suggests that specific visual features like colour, shape, or layout made distractors stand out early in the search. Feature Integration Theory (FIT) explains this through a pre-attentive stage where basic features automatically capture attention before focused processing occurs. Earlier distractor fixations reflect this initial feature-based attention, while longer target fixation durations suggest participants spent more time on cognitive verification once localized. The Friedman test confirmed target fixation

durations varied across the 13 stimuli, showing that layout and target visibility influenced visual demand. Overall, visual attention was driven by both AOI type and stimulus characteristics.

The advanced analysis also showed that visual search behaviour varied between participants, not just between AOI types or stimuli. Through K-Means clustering, participants were grouped into Fast and Slow Searchers, suggesting that some participants were able to locate targets more efficiently, while others needed a longer and more effortful search process. This supports the idea that visual search is influenced by both the visual features of the stimulus and individual differences in attentional control. The Random Forest result further strengthened this interpretation, as the searcher profiles could be predicted from fixation-based features with high accuracy. This suggests that the clusters reflected meaningful behavioural patterns rather than arbitrary groupings. Practically, these findings imply that visual interfaces should reduce unnecessary distractors while also recognising that users may differ in how quickly they locate and verify important information.

## **7. LIMITATIONS AND SUGGESTION FOR FUTURE WORK**

Several limitations affect these findings. First, zero-fixation values do not confirm a participant missed the target, as peripheral vision might have been used. Second, missing TTFF values for unfixed AOIs restricted some statistical analyses. K-Means clustering results remain exploratory rather than definitive behavioral categories. Additionally, the study lacked direct behavioral measures, such as accuracy and response time, relying instead on passive eye-tracking. Future research should integrate behavioral metrics with eye-tracking and utilize raw gaze data for detailed scanpath analysis. Testing clustering results on larger datasets will also verify the consistency of searcher profiles across different participants and stimuli.

## **8. CONCLUSION**

Our analysis of 130 participants reveals visual attention in real-world environments is a balance between bottom-up distraction and top-down focus. While distractors capture early attention faster, targets require deeper cognitive processing, leading to longer fixation durations. Significant performance variations across the 13 stimulus images highlight the influence of background complexity on visual search. K-means clustering identified distinct Fast and Slow searcher profiles, a classification validated by an 87%

accurate Random Forest model. These findings support Feature Integration Theory in ecologically valid settings and offer data-driven UI/UX guidelines for designing interfaces aligned with human scanning patterns and cognitive speeds.

## 9. ACKNOWLEDGEMENT

The authors express their gratitude to the Faculty of Computer Science and Information Technology (FCSIT), University Malaya, for providing the necessary infrastructure and access to the screen-based Tobii eye-tracking system in BT2. We also thank the 130 undergraduate students who volunteered their time to participate in this data collection process.

### **Artificial Intelligence (AI) Use Disclaimer**

Following institutional academic integrity guidelines, an artificial intelligence tool was responsibly used during the final finalization of this report. Specifically, AI assistance was limited to structuring text layouts, improving sentence readability, and proofreading the English manuscript for clarity. The core components of this study, including the Python data preprocessing scripts, inferential statistical analyses ( $t$ -tests and Friedman tests), machine learning configurations (K-means and Random Forest models), and the cognitive interpretations of findings, were conducted, checked, and verified by the student researchers. AI assistance was also used to cross-check whether the selected analytical approaches were clearly aligned with the research questions and project requirements. The final decisions on the methods, results, and interpretations remained the responsibility of the student researchers.

## 10. REFERENCES

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## 11. GANTT CHART AND TASK DISTRIBUTIONS

| Task                                                                    | Members  | Week Start | Week Due |
|-------------------------------------------------------------------------|----------|------------|----------|
| Tobii Setup + AOI + Export                                              | Izzat    | Week 9     | Week 10  |
| Introduction + Gap / Problem / Challenge                                | Aqil     | Week 10    | Week 11  |
| Research Questions + Objectives + Hypotheses + Contribution             | Olindo   | Week 10    | Week 11  |
| Literature Review                                                       | Darwisyi | Week 11    | Week 12  |
| Methodology: Participants + Materials                                   | Rahinul  | Week 11    | Week 12  |
| Methodology: Procedure + Experimental Design                            | Irfan    | Week 11    | Week 12  |
| Data Preprocessing + Python Analysis Workflow                           | Fakhrul  | Week 10    | Week 12  |
| Advanced Analysis                                                       | Edlan    | Week 11    | Week 12  |
| Descriptive Results                                                     | Akram    | Week 11    | Week 12  |
| Statistical Results + Discussion + Limitations                          | Mukhlis  | Week 12    | Week 13  |
| Abstract + Conclusion + Acknowledgement + References + Final Formatting | Najla    | Week 12    | Week 13  |

Table 5: Gantt chart & Task distribution